Chapter 14: Further Conjugation Results

Convex Analysis and Monotone Operator Theory in Hilbert Spaces

H.H. Bauschke and P.L. Combettes

2e, 2017

14.1: Strongly Convex Conjugates

Key question: When does the conjugate of a function relate to strong convexity?

Proposition (14.2)

Let $f : \mathcal{H} \to \mathbb{R} \cup \{+\infty\}$ be continuous and convex, let $\gamma \in \mathbb{R}_{++}$, and set $q = (1/2)\|\cdot\|^2$. Then:

(i) *$f^* - \gamma^{-1}q$ is convex, i.e., f^* is γ^{-1}-strongly convex.*

(ii) *$\gamma q - f$ is convex.*

Key insight: Strong convexity of f^* is equivalent to $\gamma q - f$ being convex.

Strongly Convex Conjugates (cont'd)

Theorem (14.3)

Let $f \in \Gamma_0(\mathcal{H})$ and $\gamma \in \mathbb{R}_{++}$. Set $q = (1/2)\|\cdot\|^2$. Then:

(i) $\gamma^{-1} q = f \square (\gamma^{-1} q) + (f^* \square (\gamma q)) \circ \gamma^{-1} Id$

(ii) $Id = \mathrm{Prox}_f + \gamma\, \mathrm{Prox}_{f^*/\gamma} \circ \gamma^{-1} Id$

(iii) *For $x \in \mathcal{H}$:*

$$f(\mathrm{Prox}_f x) + f^*(\mathrm{Prox}_{f^*/\gamma}(x/\gamma)) = \langle \mathrm{Prox}_f x \mid \mathrm{Prox}_{f^*/\gamma}(x/\gamma) \rangle$$

where $\square$ denotes infimal convolution (Definition 12.34).

Remark 14.4: Moreau's Decomposition

When $f = \iota_K$ (indicator of closed convex cone K) with $q = (1/2)\|\cdot\|^2$:

$$(f \square q) + (f^* \square q) = q \quad \text{and} \quad \operatorname{Prox}_f + \operatorname{Prox}_{f^*} = \operatorname{Id}$$

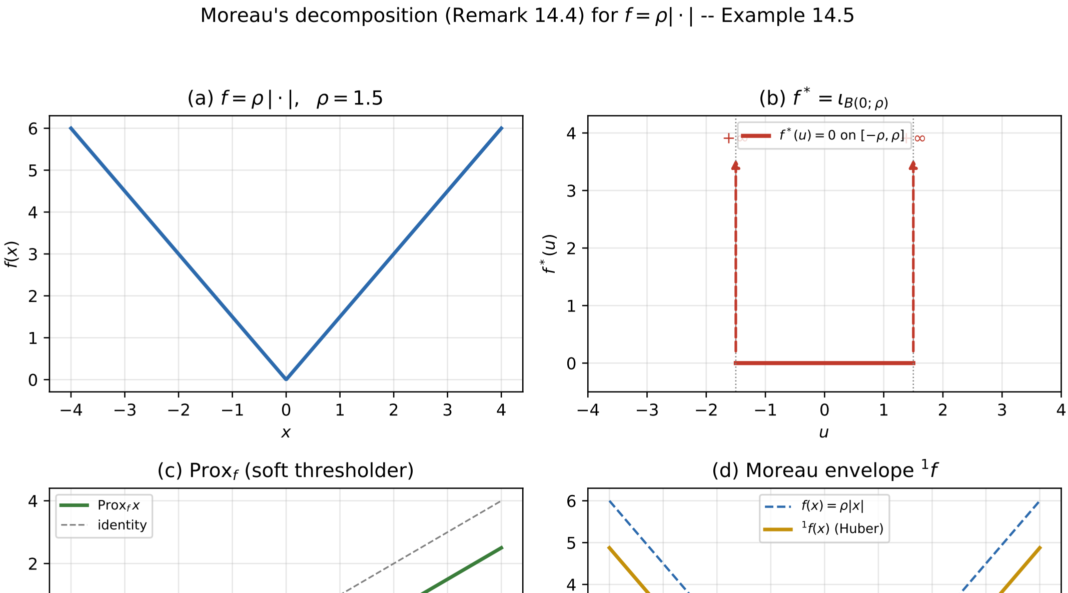
This recovers **Moreau's conical decomposition** (Theorem 6.30).

Example (14.5)

For $f(x) = \rho\|x\|$ (soft threshold function):

- $f^* = \iota_{B(0;\rho)}$ *(indicator of ball)*
- $\operatorname{Prox}_f x = \begin{cases} (1 - \rho/\|x\|)\, x & \text{if } \|x\| > \rho \\ 0 & \text{if } \|x\| \le \rho \end{cases}$ *(soft thresholder)*
- ${}^{\gamma}f(x) = \begin{cases} \rho\|x\| - \rho^2/2 & \text{if } \|x\| > \rho \\ \|x\|^2/2 & \text{if } \|x\| \le \rho \end{cases}$ *(Huber function)*

Example 14.5 Visualization



14.2: Proximal Average

Definition (14.6)

Let $f, g \in \Gamma_0(\mathcal{H})$*. The* **proximal average** *of* f *and* g *is:*

$$\mathrm{pav}(f,g) : \mathcal{H} \to \mathbb{R} \cup \{+\infty\}, \quad x \mapsto \frac{1}{2} \inf_{(y,z)\in\mathcal{H}\times\mathcal{H},\, y+z=2x} \left(f(y) + g(z) + \frac{1}{4}\|y-z\|^2 \right)$$

Proposition (14.7)

Set $L(y,z) = (y+z)/2$ *and* $F(y,z) = \frac{1}{2}f(y) + \frac{1}{2}g(z) + \frac{1}{8}\|y-z\|^2$*. Then for* $f, g \in \Gamma_0(\mathcal{H})$*:*

(i) $\mathrm{pav}(f,g) = L \triangleright F$

(ii) $\mathrm{pav}(f,g) = L \triangleright F$

(iii) $\mathrm{dom}\,\mathrm{pav}(f,g) = \frac{1}{2}\,\mathrm{dom}\, f + \frac{1}{2}\,\mathrm{dom}\, g$

(iv) $\mathrm{pav}(f,g)$ *is a proper convex function*

Proximal Average Properties

Corollary (14.8)

Let $f, g \in \Gamma_0(\mathcal{H})$ *and* $q = (1/2)\|\cdot\|^2$*. Then:*

(i) $\mathrm{pav}(f, g) \in \Gamma_0(\mathcal{H})$

(ii) $(\mathrm{pav}(f, g))^* = \mathrm{pav}(f^*, g^*)$ *(conjugate of proximal average)*

(iii) $\mathrm{pav}(f, g) \square q = \frac{1}{2}(f \square q) + \frac{1}{2}(g \square q)$

(iv) $\mathrm{Prox}_{\mathrm{pav}(f,g)} = \frac{1}{2}\mathrm{Prox}_f + \frac{1}{2}\mathrm{Prox}_g$

Remark: The proximal average is self-dual (part ii) and satisfies a beautiful sandwich property (Proposition 14.9).

Proximal Average Bounds

Proposition (14.9)

Let $f, g \in \Gamma_0(\mathcal{H})$. *Then:*

$$\left(\frac{1}{2}f^* + \frac{1}{2}g^*\right)^* \leq \operatorname{pav}(f, g) \leq \frac{1}{2}f + \frac{1}{2}g$$

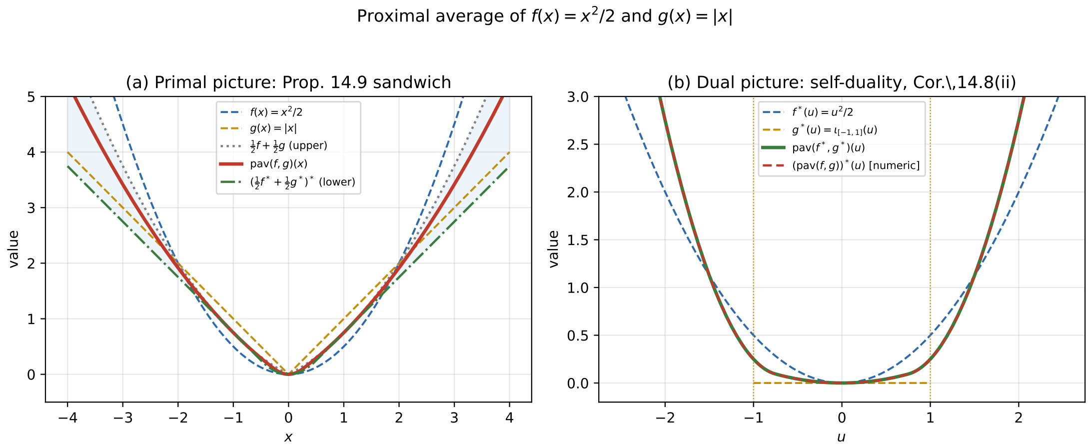
Proposition (14.10)

Let $F, G \in \Gamma_0(\mathcal{H} \oplus \mathcal{H})$. *Then:*

$$(\operatorname{pav}(F, G))^\top = \operatorname{pav}(F^\top, G^\top)$$

where $F^\top(y, z) = F(z, y)$ *is the transpose.*

Proximal Average Visualization



14.3: Positively Homogeneous Functions

Proposition (14.11)

Let $f : \mathcal{H} \to \mathbb{R} \cup \{+\infty\}$ *and set* $C = \{u \in \mathcal{H} \mid (\forall x \in \mathcal{H})\langle x \mid u\rangle \leq f(x)\}$.
Then the following are equivalent:

(i) f *is positively homogeneous and* $f \in \Gamma_0(\mathcal{H})$

(ii) $f = \sigma_C$ *and* C *is nonempty, closed, and convex*

(iii) f *is the support function of a nonempty closed convex subset of* $\mathcal{H}$

Key insight: Positively homogeneous convex functions are exactly support functions of closed convex sets.

Support Function and Minkowski Gauge

Proposition (14.12)

Let C be a convex subset of $\mathcal{H}$ such that $0 \in C$. Then:

$$m_C^* = \iota_{C^\circ}$$

where m_C is the Minkowski gauge of C.

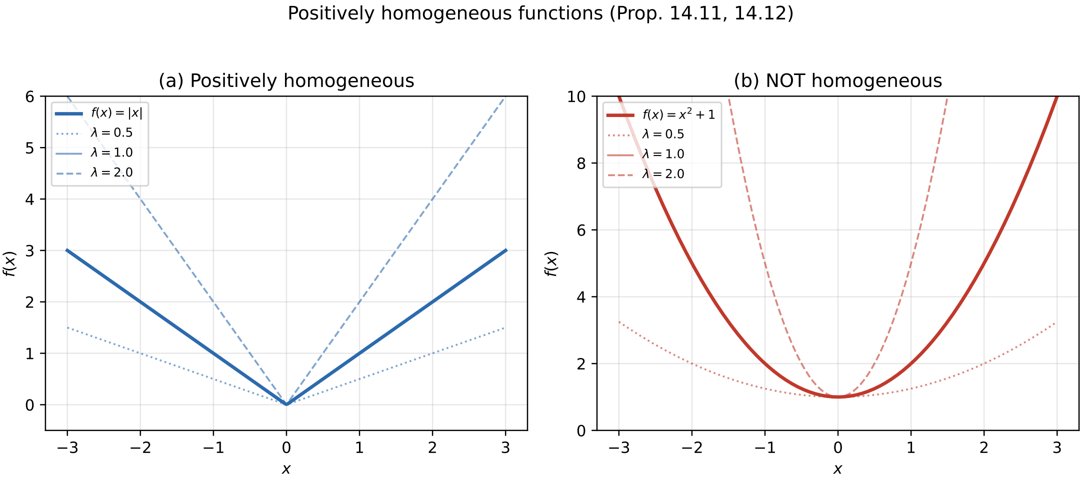
Corollary (14.13)

Let C be a closed convex subset of $\mathcal{H}$ such that $0 \in C$. Then:

(i) $C = lev_{\leq 1} m_C$

(ii) *If* $0 \in intC$, *then* $intC = lev_{<1} m_C$

Positively Homogeneous Functions Visualization



Proposition (14.14)

Let $f \in \Gamma_0(\mathcal{H})$, $\alpha \in \mathbb{R}_{++}$, *and consider:*

(i) $\liminf_{\|x\| \to +\infty} f(x)/\|x\| > \alpha$

(ii) $(\exists \beta \in \mathbb{R})\ f \geq \alpha \|\cdot\| + \beta$

(iii) $(\exists \gamma \in \mathbb{R})\ f^*|_{B(0,\alpha)} \leq \gamma$

(iv) $\liminf_{\|x\| \to +\infty} f(x)/\|x\| \geq \alpha$

Then (*i*) $\Rightarrow$ (*ii*) $\Leftrightarrow$ (*iii*) $\Rightarrow$ (*iv*).

Characterization of Coercivity

Proposition (14.15)

Let $f \in \Gamma_0(\mathcal{H})$. *Consider:*

(i) f *is supercoercive*

(ii) f^* *is bounded on every bounded subset of* $\mathcal{H}$

(iii) $\operatorname{dom} f^* = \mathcal{H}$

Then $(i) \Leftrightarrow (ii) \Rightarrow (iii)$.

Theorem (14.16)

Let $f \in \Gamma_0(\mathcal{H})$. *The following are equivalent:*

(i) f *is coercive*

(ii) *The lower level sets* $(\mathrm{lev}_{\leq c} f)_{c \in \mathbb{R}}$ *are bounded*

(iii) $\liminf_{\|x\| \to +\infty} f(x)/\|x\| > 0$

v)-(vi) *[Additional characterizations via subdifferentials]*

Moreau-Rockafellar Theorem

Theorem (14.17: Moreau-Rockafellar)

Let $f \in \Gamma_0(\mathcal{H})$ and $u \in \mathcal{H}$. Then:

$$f - \langle \cdot \mid u \rangle \text{ is coercive} \quad \Leftrightarrow \quad u \in \operatorname{int} \operatorname{dom} f^*$$

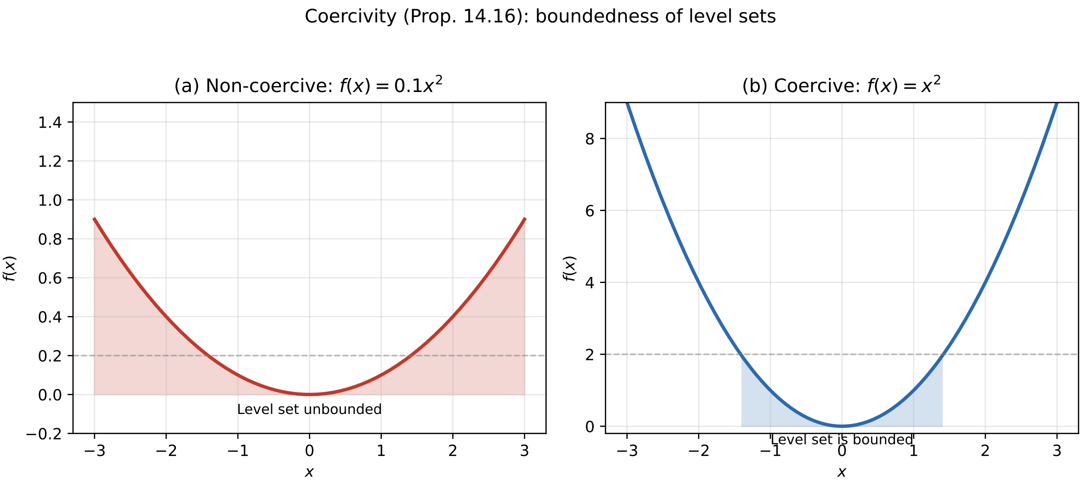
Corollary (14.18)

Let $f, g \in \Gamma_0(\mathcal{H})$ and suppose f is supercoercive. Then:

(i) *$f \Box g$ is coercive if and only if g is coercive*

(ii) *$f \Box g$ is supercoercive if and only if g is supercoercive*

Coercivity Visualization



Proposition (14.19)

Let $g : \mathcal{H} \to \mathbb{R} \cup \{+\infty\}$ *be proper, let* $h \in \Gamma_0(\mathcal{H})$*, and set:*

$$f(x) = \begin{cases} g(x) - h(x) & \text{if } x \in \operatorname{dom} g \\ +\infty & \text{if } x \notin \operatorname{dom} g \end{cases}$$

Then:

$$(\forall u \in \mathcal{H}) \quad f^*(u) = \sup_{v \in \operatorname{dom} h^*} \left(g^*(u+v) - h^*(v) \right)$$

Toland-Singer Formula

Corollary (14.20: Toland-Singer)

Let $g, h \in \Gamma_0(\mathcal{H})$. *Then:*

$$\inf_{x \in \operatorname{dom} g} (g(x) - h(x)) = \inf_{v \in \operatorname{dom} h^*} (h^*(v) - g^*(v))$$

Key insight: The Toland-Singer formula provides a dual problem for minimizing differences of convex functions. This is crucial for solving optimization problems of the form:

$$\min_x (g(x) - h(x))$$

by reformulating as a dual maximization problem.

Python Implementation: Soft Thresholding

```python
import numpy as np
import scipy.optimize as opt

# Example 14.5: Soft thresholding (proximal operator of f = rho*||x||)
def soft_threshold(x, rho):
    """Compute Prox_f(x) where f(x) = rho * ||x||"""
    return np.sign(x) * np.maximum(np.abs(x) - rho, 0)

def moreau_envelope(x, rho):
    """Compute ^1 f(x) = generalized Huber function"""
    abs_x = np.abs(x)
    return np.where(abs_x > rho,
                    rho * abs_x - rho**2 / 2,
                    abs_x**2 / 2)

# Verify Moreau decomposition: f = Prox_f + Moreau_envelope
rho = 1.5
x = np.linspace(-4, 4, 100)
f = rho * np.abs(x)
prox = soft_threshold(x, rho)
envelope = moreau_envelope(x, rho)

# Check: f(Prox_f(x)) + envelope(x) = (1/2) ||x||^2
verification = f[np.abs(prox) > 0] + envelope[np.abs(prox) > 0]
```

Proximal Average Computation

```
from scipy.optimize import minimize_scalar

def proximal_average(f, g, x, max_iter=1000):
    """
    Compute pav(f,g)(x) from Definition 14.6
    pav(f,g)(x) = (1/2) * min_{y,z: y+z=2x}
                  [f(y) + g(z) + (1/4)||y-z||^2]
    """
    def objective(y):
        z = 2*x - y
        return f(y) + g(z) + 0.25 * np.linalg.norm(y - z)**2

    result = minimize_scalar(objective, bounds=(x-10, x+10),
                             method='bounded',
                             options={'xatol': 1e-10})
    return 0.5 * result.fun

# Example: pav of f(x) = x^2/2 and g(x) = |x|
f = lambda y: 0.5 * y**2
g = lambda z: np.abs(z)

x_vals = np.linspace(-2, 2, 50)
pav_vals = [proximal_average(f, g, x) for x in x_vals]
```

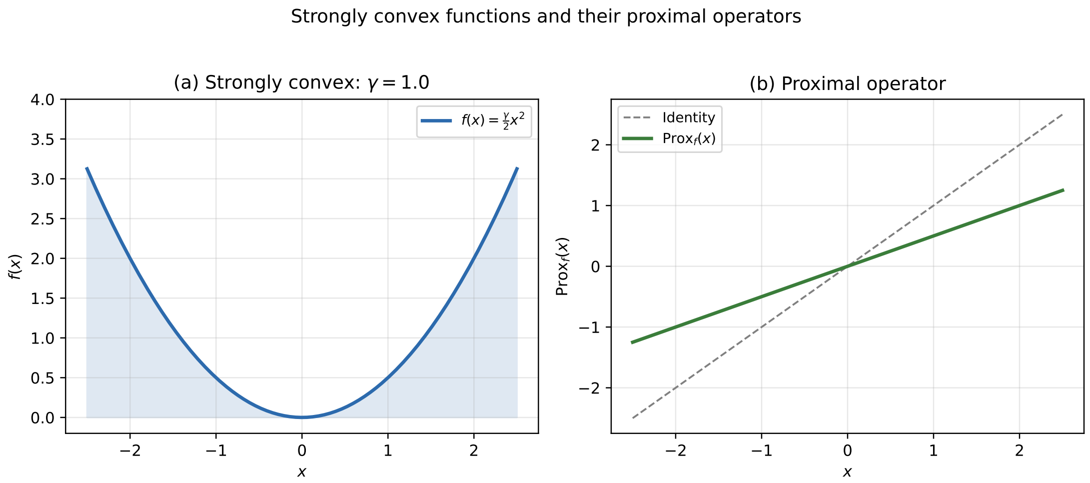
Coercivity Check

```python
def is_coercive(f, x_test=None):
    """
    Check coercivity by verifying boundedness of level sets
    (Theorem 14.16): f is coercive iff all lev_{<=c}(f) are bounded
    """
    if x_test is None:
        x_test = np.logspace(0, 3, 100)  # Test x from 1 to 1000

    # Check if f(x)/||x|| -> infinity as ||x|| -> infinity
    ratios = []
    for x in x_test:
        x_vec = np.ones(10) * x  # Test in R^10
        f_val = f(x_vec)
        x_norm = np.linalg.norm(x_vec)
        if x_norm > 0:
            ratios.append(f_val / x_norm)

    return min(ratios) > 0  # Coercive if min ratio > 0

# Example: f(x) = x^2 is coercive
f_coercive = lambda x: np.sum(x**2)

# Example: f(x) = 0.1*x^2 grows slowly, not coercive
f_noncoercive = lambda x: 0.1 * np.sum(x**2)

print(f"f(x)=x^2 coercive: {is_coercive(f_coercive)}")
print(f"f(x)=0.1*x^2 coercive: {is_coercive(f_noncoercive)}")
```

Strongly Convex Conjugates

```python
def strongly_convex_conjugate_example():
    """
    Proposition 14.2 & Theorem 14.3:
    If f is strongly convex, then f^* is strictly convex.
    """
    gamma = 1.0
    q = lambda x: 0.5 * np.linalg.norm(x)**2  # q(x) = (1/2)||x||^2

    # Example: f(x) = (gamma/2)*||x||^2, self-conjugate
    f = lambda x: (gamma / 2) * np.linalg.norm(x)**2
    f_star = lambda u: (0.5 / gamma) * np.linalg.norm(u)**2

    # Proximal operator (Theorem 14.3 part ii)
    def prox_f(x, step=1.0):
        return x / (1.0 + gamma * step)

    # Verify: f^* - (1/gamma)*q is convex
    # For this example, f^* = (1/2*gamma)*||.||^2, so
    # f^* - (1/gamma)*q = (1/2*gamma - 1/gamma)*||.||^2
    #                   = -(1/2*gamma)*||.||^2, still convex!

    x_test = np.linspace(-2, 2, 100)

    return f, f_star, prox_f
```

Strongly Convex Functions Visualization



Chapter 14: Key Takeaways

1. **Strongly Convex Conjugates** (14.1-14.2):
   - Strong convexity of f^* characterized by convexity of $\gamma q - f$
   - Theorem 14.3 connects proximal operators and conjugates
2. **Proximal Average** (14.2):
   - Midpoint between convex functions, sandwiched between $(f + g)/2$ and lower bound
   - Self-dual structure: $(\mathrm{pav}(f, g))^* = \mathrm{pav}(f^*, g^*)$
   - Proximal operator: $\mathrm{Prox}_{\mathrm{pav}(f,g)} = \frac{1}{2}\,\mathrm{Prox}_f + \frac{1}{2}\,\mathrm{Prox}_g$
3. **Positively Homogeneous Functions** (14.3):
   - Characterized as support functions of closed convex sets
   - Connection to Minkowski gauge
4. **Coercivity** (14.4):
   - Functions with bounded level sets
   - Moreau-Rockafellar theorem: $u \in \operatorname{int}\operatorname{dom} f^* \Leftrightarrow$ coercivity
5. **Conjugate of Difference** (14.5):
   - Toland-Singer formula for dual optimization

Further Reading

- **Related Chapters:**
  - Chapter 12: Duality and conjugacy fundamentals
  - Chapter 13: Subdifferentials and proximity operators
  - Chapter 15: Fenchel-Rockafellar duality
- **Key Authors & References:**
  - J.J. Moreau: Proximal analysis and decomposition theorems
  - R.T. Rockafellar: Convex analysis and duality theory
  - I. Singer, P. Tseng, H. Bauschke: Further conjugacy results
- **Applications:**
  - Soft thresholding in signal processing and compressed sensing
  - Proximal splitting algorithms (Douglas-Rachford, ADMM)
  - Machine learning: regularized convex optimization